# Unit 3 Targeted Practice — Equations & Inequalities

Complete independently · One/Two-Step · Distributive · Multi-Step · Special Cases · Inequalities · Word

## A. One- & Two-Step Equations

**Problem 1** Score 3

Solve: $5x = 45$

**Answer:** _____

**Problem 2** Score 3

Solve: $2x - 7 = 9$

**Answer:** _____

**Problem 3** Score 3

Solve: $\frac{x}{4} + 5 = 8$

**Answer:** _____

**Problem 4** Score 4

Solve: $-3x + 2 = -10$

**Answer:** _____

## B. Distributive Property & Variables on Both Sides

**Problem 1** Score 3

Solve: $2(3x + 1) = 26$

**Answer:** _____

**Problem 2** Score 4

Solve: $8x - 1 = 5x + 17$

**Answer:** _____

## C. Multi-Step Equations

**Problem 1** Score 4

Solve: $2(x + 5) = 3x - 1$

**Answer:** _____

**Problem 2** Score 4

Solve: $10 - x = 2x + 1$

**Answer:** _____

**Problem 3** Score 5

Solve: $6x + 2 = 3(x + 8)$

**Answer:** _____

**Problem 4** Score 5

Solve: $\frac{3}{4}(x + 8) - 1 = x + 2$

**Answer:** _____

## D. Special Cases — No Solution or Infinite Solutions

**Problem 1** Score 4

Solve: $3x + 7 = 3x - 4$ — how many solutions?

**Answer:** _____

**Problem 2** Score 5

Solve: 2(x + 6) = 2x + 12 — how many solutions?

**Answer:** _____

**Problem 3** Score 5

Why does 4(x − 1) + 2 = 4x − 2 have infinitely many solutions?

**A.** After you simplify, both sides are exactly the same: 4x − 2.
**B.** The x-terms are the same, so there is no solution.
**C.** x = 0 is the only solution.
**D.** Any equation with 4x on both sides has infinitely many solutions.

**Answer:** _____

## E. Linear Inequalities

**Problem 1** Score 3

Solve: 4x ≥ 32

**Answer:** _____

**Problem 2** Score 3

Solve: 2x + 7 ≤ 19

**Answer:** _____

**Problem 3** Score 4

Solve: −5x < 20 (careful!)

**Answer:** _____

**Problem 4** Score 5

Solve: −4x − 3 > 9 (careful!)

**Answer:** _____

## F. Word Problems

**Problem 1** Score 4

A phone plan costs $20 plus $0.05 per text. If the bill was $35, how many texts were sent?

**Answer:** _____

**Problem 2** Score 5

You have $60 and want to buy shirts that cost $12 each. What's the max number of shirts you can buy?

**Answer:** _____

**Problem 3** Score 5

Plan A: $40 to join a club plus $10 each month. Plan B: $20 each month with no joining fee. After how many months do the two plans cost the same?

**Answer:** _____